ITAI-1371 - Intro to Machine Learning

Individual Contribution Journal

Camila Ferreira da Silva

My contribution to Lab 05 was focused on handling missing values in the Age column of the Titanic dataset. I calculated the median age and used it to fill the missing values. I chose the median because it is less affected by extreme values than the mean. After completing the imputation, I checked the Age column again and verified that there were no missing values remaining.

Danh Nguyen

For Lab 5, I answered all three Knowledge Check questions. For the first question, I explained why the median is often better than the mean for imputation: the median is not pulled by outliers. I used examples from our data, such as the slightly right-skewed Age column and the very skewed Fare column. For the second question, I described how One-Hot Encoding turns categories into 1/0 columns. I explained why simply numbering the categories would create a false order, and why drop_first=True avoids multicollinearity. For the third question, I explained that a Decision Tree does not need feature scaling. It splits on thresholds within one feature at a time, and scaling does not change the order of the values.

Great Inioluwa Olayiwola

In this lab, I independently completed the end-to-end data preparation workflow on the Titanic dataset, addressing data quality issues to prepare the dataset for machine learning models. I resolved missing values in the numerical features by implementing median imputation on the Age column using modern Pandas syntax to avoid deprecation warnings. I then applied one-hot encoding via pd.get_dummies() with drop_first=True to transform categorical attributes (Sex and Embarked) into binary flags without introducing multicollinearity. Finally, I standardized the continuous numerical features (Age and Fare) using Scikit-Learn's StandardScaler to ensure zero mean and unit variance. All code cells were successfully executed, verified for reproducibility, and paired with analytical responses evaluating feature scaling and imputation strategies.

Quan Thach

I handled the one-hot encoding of the categorical columns. I used pd.get_dummies() on the Sex and Embarked columns of the Titanic dataframe and set drop_first=True so we wouldn't end up with redundant columns. That left us with Sex_male, Embarked_Q and Embarked_S, and the original text columns were dropped automatically. I printed the first few rows to confirm the new columns looked right, and I saved the result as encoded_df so the scaling step in Task 3 could build directly on it.